8.033 Fall 2025: Solution Set 3

1

a. In Alice's frame, light travels vertically up for a distance L to reach the second mirror, so the time that it takes is simply

$$\Delta t_A = L/c. \tag{1}$$

b. In Bob's frame, both mirror 1 and 2 are moving along with Alice at speed v_{AB}. Suppose Alice is moving in the $+x$ direction, and it takes light time Δt_B to reach mirror 2 in Bob's frame. Then notice that when light reaches mirror 2, it has already moved forward by a distance of $v_{AB}\Delta t_B$ compared to where it was when light was emitted from mirror 1 at time Δt_B ago (see Fig. 1). The distance traveled by light is simply $c\Delta t_B$. The geometry in Fig. 1 tells us

$$L^2 + v_{AB}^2 \Delta t_B^2 = c^2 \Delta t_B^2. \tag{2}$$

Solve for Δt_B, we get

$$\begin{aligned} \Delta t_B &= \frac{1}{\sqrt{1 - v_{BA}^2/c^2}} \frac{L}{c} \\ &= \frac{1}{\sqrt{1 - v_{BA}^2/c^2}} \Delta t_A, \end{aligned} \tag{3}$$

where we have used $v_{AB} = -v_{BA}$ and Eq. (1). Notice that $\frac{1}{\sqrt{1-v_{BA}^2/c^2}} = \gamma(v_{BA})$, so Eq. (3) becomes

$$\Delta t_B = \gamma(v_{BA}) \Delta t_A, \tag{4}$$

which is exactly time dilation.

Note that Bob can only directly measure the time that light takes to return to mirror 1. From the geometry shown in Fig. 1, the time for the light to return to mirror 1 is just

$$2\Delta t_B = \frac{2}{\sqrt{1 - v_{BA}^2/c^2}} \frac{L}{c} = \frac{1}{\sqrt{1 - v_{BA}^2/c^2}} (2\Delta t_A), \tag{5}$$

where we see that time dilation Eq. (4) still holds since in Alice's frame, light takes $2\Delta t_A$ to come back to mirror 1.

2

a. Since a muon decays after $10^{-6}s$, it must take the muon less than $10^{-6}s$ to reach the surface of earth in order to be directly detected. In a naive Galilean calculation, this tells us that the muon must travel through the atmosphere at a speed of

$$v = h/t = 30,000m/10^{-6}s = 3 \cdot 10^{10} m/s = 100c, \tag{6}$$

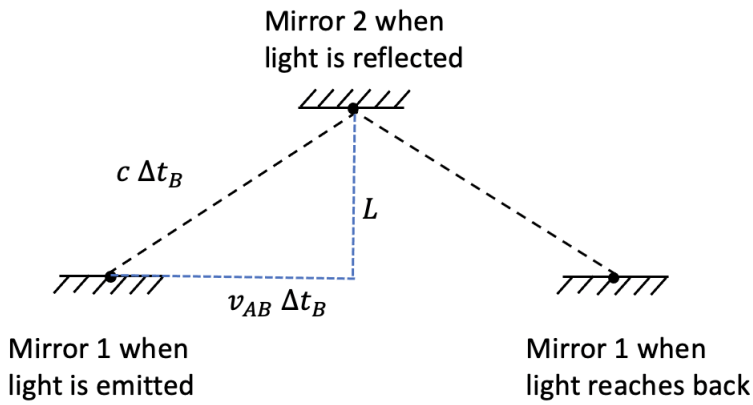

Figure 1: The positions of mirror 1 in Bob's frame when light was emitted and received back, and the position of mirror 2 in Bob's frame when light is reflected. The black dashed lines are trajectories of light.

which is just 100 in natural units. This is clearly wrong in relativity since it would indicate that muons travel at least 100 times faster than light!

b. It takes the muon $t_{decay} = 10^{-6}s$ to decay in its rest frame. When the muon is traveling at a relativitic speed, its lifetime measured on the surface of earth, t'_{decay}, is much longer due to time dilation:

$$t'_{decay} = \gamma(v_{muon})t_{decay}. \tag{7}$$

Then if a muon reaches the surface of earth before it decays, it needs to travel at least with a speed of

$$v_{muon} = h/t'_{decay}. \tag{8}$$

Plug in Eq (7) and solve for v_{muon}, we get

$$v_{muon} = \frac{h/t_{decay}}{\sqrt{1 + h^2/t^2_{decay}}} = 0.99995. \tag{9}$$

c. We know a muon has a lifetime of $t_{decay} = 10^{-6}s$. Then the time for the muon to travel down to the surface of earth, being kept at the surface, and traveling back to the atmosphere must add up to t_{decay} in the muon's rest frame. The time for the muon to travel down to earth, measured by an observer at the surface of earth, is

$$t' = h/v. \tag{10}$$

It is related to the time measured in the muon's rest frame (call it $\Delta\tau$) by

$$t' = \gamma(v)\Delta\tau. \tag{11}$$

Using Eq. (10) and (11), we can solve for $\Delta\tau$ and get

$$\Delta\tau = \frac{h}{v}\gamma^{-1}(v) = \frac{h}{v}\sqrt{1-v^2} = 4.47 \cdot 10^{-7}s. \tag{12}$$

When we keep the muon at the surface of earth, there is no time dilation effect since the muon is at rest, and the time for the muon to travel back to the atmosphere is again $\Delta\tau$. We then find the time to keep the muon at the surface of earth to be

$$\Delta t = t_{decay} - 2\Delta\tau = 1.1 \cdot 10^{-7}s. \tag{13}$$

3

Recall from the lecture that for a moving object, only the length parallel to the direction of motion is Lorentz contracted. In frame R, the length of the rod projected on the x axis and y axis are

$$\begin{aligned} L_x &= L_0 \cos\theta \\ L_y &= L_0 \sin\theta, \end{aligned} \tag{14}$$

respectively. In frame R', since the rod is moving along the x direction, only its length along the x direction is contracted, but not the length along the y direction. We thus measure the length of the rod along the x' and y' axis in frame R' to be

$$\begin{aligned} L'_x &= L_x/\gamma(v) = \sqrt{1-v^2}L_0\cos\theta \\ L'_y &= L_y = L_0 \sin\theta. \end{aligned} \tag{15}$$

Then the length of the rod in frame R' is

$$L' = \sqrt{(L'_x)^2 + (L'_y)^2} = \sqrt{(1-v^2)L_0^2\cos^2\theta + L_0^2\sin^2\theta} = L_0\sqrt{1-v^2\cos^2\theta}, \tag{16}$$

and the angle it makes with the x' axis is

$$\theta' = \tan^{-1}\frac{L'_y}{L'_x} = \tan^{-1}\frac{\sin\theta}{\sqrt{1-v^2}\cos\theta}. \tag{17}$$

4

When the camera takes a picture at time $t = 0$, it collects all the photons that reach the camera at that time. There is a finite distance between the rod and the camera, and the speed of light is also finite, so photons that reach the camera when the image is taken must have left the rod at times earlier than $t = 0$ (See Fig (2)). This means that even though the rod was located between point A' and B' at $t = 0$, photons from the trailing end of the rod that reach the camera at $t = 0$ were actually emitted at an earlier time, say $t = -|t_A|$, when the trailing end of the rod was at point A. Similarly, photons from the leading end of the

rod that reach the camera at $t = 0$ were actually emitted at an earlier time $t = -|t_B|$, when the leading end of the rod was at point B.

Examining the triangle geometry shown in Fig (2), we can find the time $|t_A|$ by noticing that it is the time for both the photons from point A to reach the camera (located at point C) and the rod to move from point A to A', so

$$|t_A|^2 = (v|t_A| + L/\gamma)^2 + h^2, \tag{18}$$

where $\gamma = 1/\sqrt{1-v^2}$, and we have set $c = 1$ and realized that the length of the rod is contracted in the frame where the camera is stationary (so the distance from point A' to the origin, point O, is L/γ instead of L). Massaging Eq. (18) into the standard form of a quadratic equation in $|t_A|$, we get

$$(v^2-1)|t_A|^2 + (2vL/\gamma)|t_A| + L^2/\gamma^2 + h^2 = 0. \tag{19}$$

Using the quadratic formula, we solve for $|t_A|$ and get

$$\begin{aligned} |t_A| &= \frac{-(2vL/\gamma) \pm \sqrt{4v^2L^2/\gamma - 4(v^2-1)(L^2/\gamma^2 + h^2)}}{2(v^2-1)} \\ &= \gamma(vL \pm \sqrt{L^2+h^2}), \end{aligned} \tag{20}$$

where we simplified the first line of Eq. (20) using $\gamma = 1/\sqrt{1-v^2}$ a few times to get the second line. Note that we need to discard the minus in Eq. (20) because $|t_A|$ is a positive quantity but $vL - \sqrt{L^2+h^2} < 0$ (since we must have $v < 1$), so $|t_A| = \gamma(vL + \sqrt{L^2+h^2})$. Then we know the trailing end of the rod at $t = -|t_A|$ (located at point A) has the position

$$x_A = -L/\gamma - v|t_A| = -\gamma(L + v\sqrt{L^2+h^2}). \tag{21}$$

This is where the trailing end of the rod appears to be in the image taken by the camera at $t = 0$.

Similarly, $|t_B|$ is the time for both photons from point B to reach the camera and the leading end of the rod to move from point B to B', so

$$|t_B|^2 = (L/\gamma - v|t_B|)^2 + h^2. \tag{22}$$

Again, using the quadratic formula and simplify, we find

$$|t_B| = \gamma(\sqrt{L^2+h^2} - vL), \tag{23}$$

where we have kept the solution such that $|t_B| > 0$. Then we know the leading end of the rod at $t = -|t_B|$ is located at

$$x_B = L/\gamma - v|t_B| = \gamma(L - v\sqrt{L^2+h^2}), \tag{24}$$

which is where the leading end of the rod appears to be in the image. The length of the rod in the image is then

$$L_{image} = x_B - x_A = 2\gamma L, \tag{25}$$

which is the proper length *elongated* by a Lorentz factor!

This might be a little puzzling at first. The key point to realize is that the photons that arrive at the camera when the image is taken did not leave the rod at the same time. In particular, we see that

$$\Delta t = |t_A| - |t_B| = 2\gamma v L > 0. \tag{26}$$

In other words, light from the trailing end of the rod left earlier than light from the leading end of the rod. During the time Δt, the rod had moved forward by $\Delta x = v\Delta t = 2\gamma v^2 L$, so the length of the rod appears to be

$$2L/\gamma + \Delta x = 2L\sqrt{1-v^2} + \frac{2v^2 L}{\sqrt{1-v^2}} = \frac{2L(1-v^2)}{\sqrt{1-v^2}} + \frac{2v^2 L}{\sqrt{1-v^2}} = 2\gamma L. \tag{27}$$

It is a known fact that a moving 3-dimensional sphere does not appear contracted in an image. More accurately, the 2-dimensional outline of a moving 3-dimensional sphere in an image has the same outline as that of a stationary sphere. In other words, the 2-dimensional outline of a moving sphere does not appear contracted or elongated in an image. Why the outline of the sphere does not appear elongated like the moving rod is a different (and much harder) problem. What we have learned from this problem, however, can conceptually explain why the sphere does not appear to be Lorentz contracted: the visual effect caused by light from different parts of the sphere arriving at the camera at different times works directly in opposition to the Lorentz contraction, which makes the sphere appear to be not contracted.

5

a. Let's call the outbound journey trip 1 and the inbound journey trip 2, then $v_1 = 0.9c$ and $v_2 = 0.99c$. We know that the time passed on earth during trip 1 and 2 are just

$$t_1' = d/v_1, \tag{28}$$
$$t_2' = d/v_2, \tag{29}$$

where d is the distance between earth and the planet. Due to time dilation, the time passed on the spaceship is different and is given by

$$t_1 = t_1'/\gamma(v_1) = \sqrt{1-\left(\frac{v_1}{c}\right)^2}\frac{d}{v_1}, \tag{30}$$
$$t_2 = t_2'/\gamma(v_2) = \sqrt{1-\left(\frac{v_2}{c}\right)^2}\frac{d}{v_2}. \tag{31}$$

We know that t_1 and t_2 should sum up to 4 years (5 years minus 1 year being on the planet), so can solve for the distance by setting $c = 1Ls/s$ (so $v_1 = 0.9Ls/s$ and $v_2 = 0.99Ls/s$) and converting years to seconds to get

$$t_1 + t_2 = \sqrt{1-\left(\frac{v_1}{c}\right)^2}\frac{d}{v_1} + \sqrt{1-\left(\frac{v_2}{c}\right)^2}\frac{d}{v_2} = 4 \times 3.15 \cdot 10^7 s, \tag{32}$$

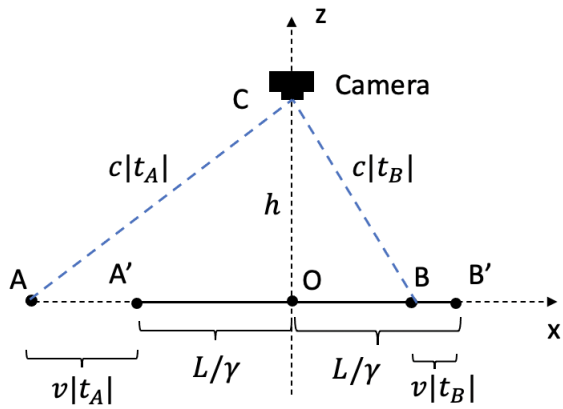

Figure 2: Diagram that illustrates problem 4. The origin is located at point O. The camera is at a distance h above the origin on the z axis. The solid black line indicates the position of the rod at $t = 0$, with its trailing and leading end located at point A' and B', respectively. However, the photons received by the camera at $t = 0$ must have left the rod before $t = 0$. This indicates that light from the trailing end of the rod that reaches the camera at $t = 0$ must have been emitted at an earlier time $t = -|t_A|$, when the trailing end of the rod was located at point A. Similarly, light from the leading end of the rod that reaches the camera at $t = 0$ must have been emitted at an early time $t = -|t_B|$, when the leading end of the rod was at point B. The blue dashed lines represent trajectories of light that reaches the camera from the rod.

from which we solve for d and find that $d = 2 \cdot 10^8 s$.

b. Using Eq. (30) and the distance we just found, we have $t_1 = 9.7 \cdot 10^7 s = 3.1 yr$.

c. In the reference frame on earth, the total time passed during the entire trip is

$$t_1' + 1yr + t_2' = \frac{d}{v_1} + 3.15 \cdot 10^7 s + \frac{d}{v_2} = 4.6 \cdot 10^8 s = 14.5 yr. \tag{33}$$

d. First, we need to find out how much time has passed on earth when the spaceship measures 30 minutes:

$$t_{earth} = t_{ship}\gamma(v_1) = 1800s \times \frac{1}{\sqrt{1 - 0.9^2}} = 4130s. \tag{34}$$

So we know that when the earth clock reads 4130 seconds, light was emitted from the spaceship. At that moment, the spaceship was

$$d_{ship} = v_1 t_{earth} = 0.9Ls/s \times 4130s = 3717Ls \tag{35}$$

away from the earth, so people on earth had to wait another 3717 seconds for light to reach them. Then the time elapsed between the moment that the rocket left and the moment that

people on earth received the signal is

$$t_{total} = t_{earth} + d_{ship}/c = 7847s = 131min. \tag{36}$$

6

a. In Alice's frame, the light signals are sent at an interval of $\Delta t_A = 1s$. Due to time dilation, this interval measured by Bob is then

$$\Delta t_B = \gamma(v)\Delta t_A = 1.67s. \tag{37}$$

b. The distance covered by Alice between blinks in Bob's frame is

$$\Delta x_B = v\Delta t_B = 0.8 \times 3 \cdot 10^8 m/s \times 1.67s = 4 \cdot 10^8 m = 1.33Ls. \tag{38}$$

c. From part a, we know that in Bob's frame, the light signals are emitted every $1.67s$. However, each light signal has to travel a distance of Δx_b more than the previous one since Alice has moved away by the amount of Δx_b, as caulculated in part b. So, the time interval between the blinks received by Bob in his frame is

$$\Delta\tau = \Delta t_B + \Delta x_B/c = 1.67s + 1.33s = 3s. \tag{39}$$

As a side comment, notice that Eq. (39) can be written as

$$\Delta\tau = \Delta t_B + \Delta x_B/c = \Delta t_B(1 + v/c) = \Delta t_A \frac{1 + v/c}{\sqrt{1 - (v/c)^2}} = \Delta t_A \sqrt{\frac{1 + v/c}{1 - v/c}}. \tag{40}$$

Taking the inverse of Eq. (40), we get

$$f_{measured} = f_{source}\sqrt{\frac{1 - v/c}{1 + v/c}}. \tag{41}$$

This is the relativistic Doppler effect.

7

Suppose that you throw the ball along the $+x$ direction and that the wall moves toward you in the -x direction, then the ball has a velocity of $+v_{BY}$ and the wall has a velocity of $-v_{WY}$ in your reference frame (equivalently, your velocity relative to the wall is $v_{YW} = +v_{WY}$). The ball bounces back from the wall at the same speed it hits the wall, so we need to find the ball's velocity relative to the wall before the impact. Using the velocity addition formula, we get

$$v_{BW} = \frac{v_{BY} + v_{YW}}{1 + v_{BY}v_{YW}} = \frac{v_{BY} + v_{WY}}{1 + v_{BY}v_{WY}}. \tag{42}$$

Then, the velocity of the ball relative to the wall after the ball bounces back is

$$v'_{BW} = -v_{BW}. \tag{43}$$

Then, we can use the velocity addition formula again to find out the ball's velocity relative to you after it bouces back:

$$v'_{BY} = \frac{v'_{BW} - v_{WY}}{1 - v'_{BW} v_{WY}} = -\frac{v_{BW} + v_{WY}}{1 + v_{BW} v_{WY}}, \tag{44}$$

where we have used Eq. (43). Now plug in Eq. (42) and get

$$\begin{aligned} v'_{BY} &= -\left(\frac{v_{BY} + v_{WY}}{1 + v_{BY} v_{WY}} + v_{WY}\right) / \left(1 + \frac{v_{BY} + v_{WY}}{1 + v_{BY} v_{WY}} v_{WY}\right) \\ &= -\frac{v_{BY} + 2v_{WY} + v_{BY} v_{WY}^2}{1 + 2v_{BY} v_{WY} + v_{WY}^2}. \end{aligned} \tag{45}$$

Now suppose $v_{BY} \to 1$, Eq. (45) becomes

$$v'_{BY} \to -\frac{1 + 2v_{WY} + v_{WY}^2}{1 + 2v_{WY} + v_{WY}^2} = -1. \tag{46}$$

This is expected, since if you send a light pulse toward the wall and measure the speed of light that is reflected back at you, you will certainly get the velocity to be -1 as the principle of relativity tells us that must be the case.